

LANGARA, BC, Canada

WMO: 718990

Lat: 54.255N

Lon: 133.058W

Elev: 43

StdP: 100.81

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-3.7	-2.0	-10.7	1.5	-2.2	-8.1	1.9	-1.1	16.8	3.5	15.6	3.6	6.7	90	0.710

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.8	17.3	14.9	16.3	14.5	15.5	14.0	15.5	16.7	14.8	15.7	14.3	15.2	3.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.0	10.7	15.9	14.4	10.3	15.2	13.8	9.9	14.7	43.5	16.8	41.6	15.8	40.2	15.2	18.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.3	12.2	10.4	DB	-5.6	20.7	2.7	3.0	-7.6	22.9	-9.2	24.6	-10.7	26.3	-12.7	28.5	(4)
(5)			WB	-6.8	16.7	2.6	1.0	-8.7	17.4	-10.2	18.0	-11.6	18.6	-13.5	19.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	4.2	4.1	4.3	6.3	8.5	10.9	12.8	13.9	12.5	9.2	5.8	4.2	(6)
(7)		DBStd	4.10	2.77	2.21	2.39	1.70	1.65	1.55	1.23	1.21	1.44	1.85	2.54	2.61	(7)
(8)		HDD10.0	1034	181	164	177	112	51	6	0	0	1	37	125	181	(8)
(9)		HDD18.3	3741	439	398	436	361	304	223	171	138	174	284	375	439	(9)
(10)		CDD10.0	336	0	0	0	1	6	33	88	121	77	11	1	0	(10)
(11)		CDD18.3	1	0	0	0	0	0	1	0	0	0	0	0	0	(11)
(12)		CDH23.3	4	0	0	0	0	0	3	0	1	0	0	0	0	(12)
(13)		CDH26.7	1	0	0	0	0	0	1	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	4.9	5.7	5.3	5.3	4.5	4.3	3.9	3.9	3.9	4.5	5.4	5.8	5.9	(14)
(15)	Precipitation	PrecAvg	1952	197	151	147	137	106	96	90	120	186	277	225	222	(15)
(16)		PrecMax	2396	357	259	275	294	266	154	202	256	313	426	402	403	(16)
(17)		PrecMin	1475	65	18	23	54	34	37	22	25	64	132	90	75	(17)
(18)		PrecStd	223	64	56	54	50	48	35	43	58	63	68	71	63	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	9.9	10.2	13.8	15.9	18.8	17.6	19.2	17.8	14.3	11.7	10.1	(19)
(20)			MCWB	9.0	8.5	8.5	9.0	10.5	13.6	15.2	15.9	15.6	13.1	10.6	9.5	(20)
(21)		2%	DB	9.0	8.6	9.0	11.0	13.1	15.1	16.2	17.6	16.3	13.1	10.6	9.0	(21)
(22)			MCWB	8.4	7.6	7.7	8.5	10.6	12.9	14.5	15.4	14.6	11.9	9.8	8.4	(22)
(23)		5%	DB	8.2	7.7	8.1	9.7	11.8	14.0	15.4	16.6	15.4	12.3	9.7	8.3	(23)
(24)			MCWB	7.6	6.8	7.0	7.8	9.9	12.2	13.9	14.9	13.9	11.2	9.0	7.8	(24)
(25)		10%	DB	7.5	7.0	7.3	8.8	10.9	13.0	14.7	15.8	14.7	11.6	9.1	7.6	(25)
(26)			MCWB	6.9	6.1	6.3	7.2	9.3	11.5	13.5	14.4	13.5	10.6	8.4	7.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.5	9.1	9.2	10.3	12.2	14.2	15.6	16.7	16.4	13.8	11.1	9.7	(27)
(28)			MCDB	9.6	9.5	9.7	12.1	13.4	16.8	16.8	18.0	17.1	14.2	11.3	9.9	(28)
(29)		2%	WB	8.5	7.8	8.1	9.1	11.0	13.1	14.7	15.8	15.1	12.4	10.1	8.7	(29)
(30)			MCDB	8.8	8.3	8.7	10.4	12.4	14.6	15.7	17.0	15.8	12.8	10.4	8.9	(30)
(31)		5%	WB	7.7	7.0	7.2	8.2	10.2	12.4	14.2	15.1	14.4	11.6	9.2	8.0	(31)
(32)			MCDB	8.1	7.5	7.8	9.4	11.5	13.6	15.1	16.1	15.0	12.1	9.6	8.2	(32)
(33)		10%	WB	7.0	6.3	6.5	7.5	9.5	11.7	13.7	14.6	13.7	10.8	8.5	7.1	(33)
(34)			MCDB	7.4	6.8	7.2	8.7	10.7	12.8	14.5	15.5	14.4	11.4	9.0	7.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	2.9	3.2	3.9	4.4	4.1	3.9	3.4	3.8	3.8	3.3	3.1	2.9	(35)
(36)			MCDBR	3.3	3.6	4.5	5.4	5.5	5.4	4.4	5.0	4.6	4.0	3.4	3.2	(36)
(37)			MCWBR	3.2	3.2	3.6	3.9	3.8	3.6	3.1	3.4	3.4	3.3	3.2	3.2	(37)
(38)		5% WB	MCDBR	3.3	3.3	4.2	5.0	4.9	4.7	4.0	4.5	4.2	3.7	3.4	3.3	(38)
(39)			MCWBR	3.3	3.1	3.7	3.8	3.6	3.4	3.1	3.3	3.5	3.5	3.4	3.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.312	0.319	0.343	0.382	0.378	0.371	0.374	0.372	0.370	0.347	0.317	0.305	(40)	
(41)		taud	2.367	2.481	2.426	2.303	2.386	2.448	2.486	2.506	2.452	2.417	2.336	2.294	(41)	
(42)		Ebn at Noon	643	785	838	844	869	878	867	850	799	729	620	558	(42)	
(43)		Edh at Noon	55	68	89	114	111	105	100	93	87	72	56	48	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.51	1.30	2.38	3.82	4.66	4.94	4.43	3.85	2.69	1.48	0.70	0.40	(44)	
(45)		RadStd	0.06	0.21	0.28	0.19	0.34	0.38	0.40	0.35	0.27	0.16	0.07	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page